

CELEBAL INTERNSHIP WEEKLY REPORT

WEEK 4

Title: Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective: To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

Submitted By

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Celebal Technologies

Objective

The main goal of this week's assignment was to learn the basics of Microsoft Azure and understand how Azure Data Factory can be used to build a simple end-to-end data pipeline. Instead of writing code, I used Azure services to move a CSV file from one Blob Storage container to another.

During this assignment, I focused on:

- Learning how to navigate the Azure Portal and understand the purpose of Resource Groups and Azure resources.
- Creating a Storage Account and using Blob Containers to store and manage data files.
- Exploring Azure Data Factory and understanding its main sections: **Author**, **Monitor**, and **Manage**.
- Creating **Linked Services** and **Datasets** to connect Azure Data Factory with Blob Storage.
- Using the **Get Metadata** activity to verify that the source file exists before processing it.
- Building a **Copy Data** pipeline to copy the **Superstore.csv** file from the **raw-data** container to the **processed-data** container.
- Understanding the importance of **Azure IAM (Identity and Access Management)** and assigning the required roles so that Azure Data Factory could securely access the Storage Account.

By the end of this assignment, I was able to successfully create, execute, and monitor an end-to-end data pipeline in Azure.

1.Introduction

During **Week 4** of my **Data Engineering Internship at Celebal Technologies**, I got the opportunity to work with **Microsoft Azure** and gain hands-on experience with cloud-based data engineering tools. This was my first time using Azure services, so the assignment helped me understand how different Azure resources work together to build a complete data pipeline.

For this assignment, I used the **Superstore.csv** dataset to create an end-to-end data pipeline. The process started by uploading the dataset to **Azure Blob Storage**, then using **Azure Data Factory (ADF)** to validate the file with the **Get Metadata** activity and copy it from the **raw-data** container to the **processed-data** container using the **Copy Data** activity.

Throughout this assignment, I learned how to create and manage Azure resources such as a **Resource Group**, **Storage Account**, **Blob Containers**, and **Azure Data Factory**. I also explored how **Linked Services**, **Datasets**, and **Pipelines** work together to automate data movement without writing custom code.

Completing this project gave me a practical understanding of Azure Data Factory and the basic workflow followed in data engineering projects. It also improved my confidence in working with cloud services and building simple ETL pipelines using Microsoft Azure.

2. Azure Cloud Fundamentals

Before working on this assignment, I had only a basic idea about cloud computing. I knew that cloud platforms allow us to store data and run applications over the internet, but I had never worked with a cloud service in a practical environment. This week's tasks gave me my first hands-on experience with **Microsoft Azure**, which helped me understand how cloud resources are created and managed. Microsoft Azure is a cloud computing platform that provides various services such as virtual machines, storage, databases, networking, and analytics. Instead of purchasing and maintaining physical servers, users can create and use these resources whenever needed and pay only for what they use.

One of the first concepts I learned was how Azure organizes its resources. Everything starts with an **Azure Subscription**, which provides access to Azure services. Inside the subscription, resources are grouped using a **Resource Group**. A resource group acts as a logical container where all the resources related to a project are stored together. This makes it easier to organize, manage, and monitor the resources throughout the project.

For this assignment, I created a resource group and added all the required resources, including a **Storage Account** and an **Azure Data Factory** instance. Keeping everything inside the same resource group made resource management much simpler. It also helped me understand that deleting a resource group removes all the resources associated with that project, which is useful while working on practice assignments.

The main Azure services used in this assignment were **Azure Blob Storage** and **Azure Data Factory (ADF)**. Azure Blob Storage was used to store the CSV dataset, while Azure Data Factory was used to create and manage the data pipeline that copied the file from the source container to the destination container. Since both of these are Platform as a Service (PaaS) offerings, Azure manages the underlying infrastructure, allowing users to focus only on configuring and using the services.

Overall, this assignment gave me a clear understanding of Azure's basic architecture and helped me learn how cloud resources work together to build a simple yet effective data engineering solution.

3.Azure Storage Account

A **Storage Account** is an Azure service used to store and manage different types of data, such as files and blobs. In this assignment, I created a storage account named **stcelebalweek4arati** to store the **Superstore.csv** dataset. It was used as the source and destination for the Azure Data Factory pipeline, allowing the entire data transfer process to be completed within the same storage account.

4.Azure Blob Storage

Azure Blob Storage is used to store unstructured data such as CSV files, images, videos, and documents. In this assignment, I created two Blob containers: **raw-data** to store the original **Superstore.csv** file and **processed-data** to store the copied file after the pipeline execution. Blob Storage was used because it integrates easily with Azure Data Factory and is well suited for storing and transferring CSV files efficiently.

5.Azure Data Factory

Azure Data Factory (ADF) is a cloud-based data integration service used to create and manage data pipelines. It helps move data from one location to another without writing complex code. In this assignment, I used ADF to create a pipeline that copied the **Superstore.csv** file from the **raw-data** container to the **processed-data** container. To build the pipeline, I first created a **Linked Service** to connect ADF with Azure Blob Storage, then created **Source** and **Destination Datasets** to define the input and output files. Finally, I used the **Get Metadata** activity to validate the source file and the **Copy Data** activity to transfer the file successfully.

6.Resource Group

The **Resource Group (rg-celebal-week4-adf)** was the first Azure resource I created for this assignment. It acted as a logical container to organize all the project resources, including the **Storage Account** and **Azure Data Factory**. Keeping all related resources in one resource group made them easier to manage, monitor, and delete together after completing the assignment.

7.Storage Account

The **Storage Account** was created using **Standard** performance and **Locally Redundant Storage (LRS)**, which is suitable for learning and practice projects. Inside the storage account, I created two Blob containers: **raw-data** to store the original **Superstore.csv** file and **processed-data** to store the copied file after the pipeline execution. The dataset was uploaded to the **raw-data** container using the Azure Portal.

8.Linked Services

A **Linked Service** in Azure Data Factory is used to connect ADF with a data source or destination. In this assignment, I created a Linked Service named **LS_BlobStorage** using **Azure Blob Storage** with **Account Key authentication**. This connection was used by both the source and destination datasets, allowing Azure Data Factory to access the files stored in the same Storage Account.

9.Datasets

A **Dataset** in Azure Data Factory defines the data that will be used in the pipeline. It specifies the file location, format, and structure. For this assignment, I created two datasets: **DS_Source**, which points to the **Superstore.csv** file in the **raw-data** container, and **DS_Destination**, which points to the **processed-data** container. Both datasets use the **Delimited Text (CSV)** format with **comma (,)** as the **delimiter** and the **first row as the header**, matching the structure of the dataset.

10.Get Metadata Activity

The **Get Metadata** activity is used to check the details of a file before processing it. It verifies whether the source file exists and retrieves information such as the **file name**, **file type**, **size**, and **last modified date**. In this assignment, I configured the **Get Metadata** activity for the **DS_Source** dataset to validate the **Superstore.csv** file before the **Copy Data** activity was executed. This ensured that the source file was available and ready for processing.

11.Copy Data Activity

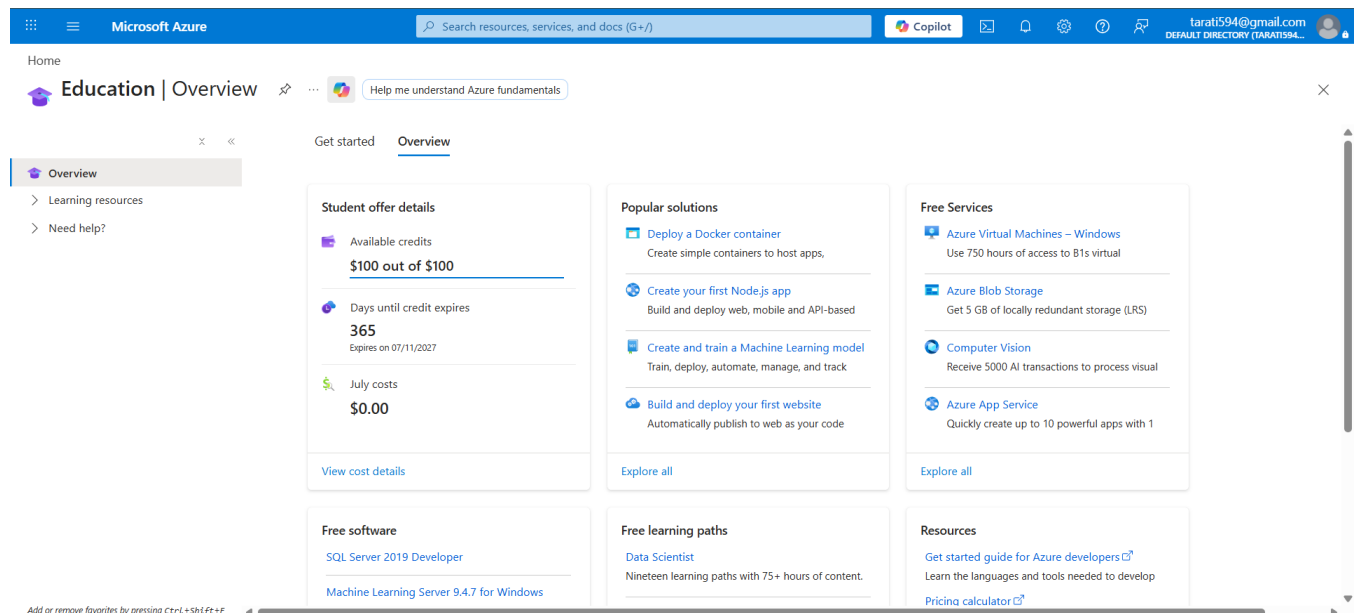
The **Copy Data** activity is used to transfer data from a source location to a destination location in Azure Data Factory. In this assignment, I configured the **DS_Source** dataset as the source and **DS_Destination** as the destination. The pipeline copied the **Superstore.csv** file from the **raw-data** container to the **processed-data** container using the default Azure Integration Runtime. Since both files had the same CSV format, the schema was mapped automatically, and the data was copied successfully.

❖ Step-by-Step Implementation

This section explains the steps I followed to complete the Azure assignment. I have described each task in the same order as it was performed, along with a brief explanation of why each step was important. This approach helped me understand not only how to use Azure services but also how they work together to build a complete data pipeline.

Task 1: Explore Azure Portal and Create a Resource Group

I logged in to the Azure Portal and explored the available services to understand the Azure environment. I created a Resource Group named **rg-celebal-week4-adf**, which acted as a logical container for all the Azure resources used in this project, including the Storage Account and Azure Data Factory. Organizing all resources in a single resource group made them easier to manage throughout the assignment.



Microsoft Azure

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Home > Resource Manager | Resource groups

Create a resource group

BasicsTagsReview + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription * Azure for Students

Resource group name * rg-celebal-week4-adf

Region * (Asia Pacific) Central India

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BasicsTagsReview + create

[Automation Link](#)

Basics

Subscription	Azure for Students
Resource group name	rg-celebal-week4-adf
Region	Central India

Tags

None

Previous

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Create

Task 2: Create a Storage Account, Blob Container, and Upload the CSV

After creating the Resource Group, I created a **Storage Account** using **Standard Performance** and **Locally Redundant Storage (LRS)**. Inside the Storage Account, I created two Blob containers:

- **raw-data** – stores the original dataset.
- **processed-data** – stores the copied output after pipeline execution.

Finally, I uploaded the **Superstore.csv** file to the **raw-data** container.

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Create a storage account

BasicsAdvancedNetworkingData protectionSecurityEncryptionTagsReview + create

View automation template

Basics

Subscription

Azure for Students

Resource group

rg-celebal-week4-adf

Location

Central India

Storage account name

stcelebalweek4arati

Primary service

Azure Blob Storage or Azure Data Lake Storage

Performance

Standard

Replication

Locally redundant storage (LRS)

Advanced

Enable hierarchical namespace

Disabled

Enable SFTP

Disabled

Enable network file system v3

Disabled

Allow cross-tenant replication

Disabled

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stcelebalweek4arati Containers

Storage account

Search

data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Containers

Classic file shares

Queues

Tables

Security + networking

Data management

Settings

Monitoring

Automation

Add or remove favorites by pressing Ctrl+Shift+F

Successfully created storage container
Successfully created storage container 'raw-data'.

+ Add container Upload Refresh Delete Change access level Restore containers Edit columns

Search containers by prefix

Only show active containers

Showing all 2 items

☐

Name

Last modified

Anonymous access level

Lease state

Storage connector

☐

logs

7/11/2026, 11:49:22 PM

Private

Available

-

☐

raw-data

7/12/2026, 12:16:08 AM

Private

Available

-

Microsoft Azure

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raw-data

Container

Search

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

+ Add Directory Upload Change access level Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

raw-data

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

☐

Name

Last modified

Access tier

Blob type

Size

Lease state

☐

Superstore.csv

7/12/2026, 12:19:31 AM

Hot (Inferred)

Block blob

2.18 MiB

Available

Add or remove favorites by pressing Ctrl+Shift+F

Successfully uploaded blob(s)
Successfully uploaded 1 blob(s).

Task 3: Create Azure Data Factory and Build Linked Services / Datasets

Next, I created an **Azure Data Factory (ADF)** instance and explored its three main sections:

- **Author** – Used to create datasets and pipelines.
- **Monitor** – Used to track pipeline execution.
- **Manage** – Used to configure Linked Services and Integration Runtime.

To connect Azure Data Factory with Blob Storage, I created a **Linked Service** named **LS_BlobStorage** using Account Key authentication.

After that, I created two datasets:

- **DS_Source** – Connected to **raw-data/Superstore.csv**
- **DS_Destination** – Connected to the **processed-data** container

I also configured the **Get Metadata** activity to verify that the source file existed before executing the pipeline.

The screenshot displays the Microsoft Azure portal interface for the 'CreateDataFactory-20260712002720' deployment. The top navigation bar shows the 'Microsoft Azure' logo and a search bar. The main content area is titled 'Overview' and features a green checkmark indicating that the deployment is complete. Key details include the deployment name, subscription (Azure for Students), resource group (rg-celebal-week4-adf), start time (7/12/2026, 12:28:48 AM), and correlation ID (feafc34b-83ef-4a54-bb82-87fbc24b8ff4). A 'Go to resource' button is visible. The left sidebar contains navigation links for Overview, Inputs, Outputs, and Template. The right sidebar provides recommendations for cost management, Microsoft Defender for Cloud, and free Microsoft tutorials.

Microsoft Azure

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Home > CreateDataFactory-20260712002720 | Overview

rg-celebal-week4-adf
Resource group

How to manage these changes more efficiently with deployment tools? Help me generate Terraform for this resource group configuration. +1

Search

CreateManage viewDelete resource groupRefreshExport to CSVOpen queryAssign tagsMoveDeleteGroup by none

Overview

Activity logAccess control (IAM)TagsResource visualizerEventsSettingsCost ManagementMonitoringAutomationHelp

Essentials

JSON View

ResourcesRecommendations

Filter for any field...Type equals allLocation equals allAdd filter

Name 1	Type	Location
adf-week4-arati	Data factory (V2)	Central India
stcelebalweek4arati	Storage account	Central India

Add or remove favorites by pressing **Ctrl+Shift+F**

Showing 1 - 2 of 2. Display count: auto

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Data Factory > adf-week4-arati

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Set up code repository

Data factory

adf-week4-arati

New

Ingest

Copy data at scale once or on a schedule.

Orchestrate

Code-free data pipelines.

Transform data

Transform your data using data flows.

Configure SSIS

Manage & run your SSIS packages in the cloud.

Recent resources

Microsoft Azure | Data Factory | adf-week4-arati

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Validate all Publish all 1

Migrate to Fabric (Preview)

Linked services

Linked service defines the connection information to a data store or compute. [Learn more](#)

+ New

Filter by name Annotations: Any

Showing 1 - 1 of 1 items

Name ↑↓	Type ↑↓	Related ↑↓	Annotations ↑↓
LS_BlobStorage	Azure Blob Storage	2	

Microsoft Azure | Data Factory | adf-week4-arati

Search

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Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Validate all Publish all 1

Migrate to Fabric (Preview)

Factory Resources

Filter resources by name

- Pipelines 0
- Change Data Capture (preview) 0
- Datasets 1
 - DS_Source
- Data flows 0
- Power Query 0

DS_Source

DelimitedText
DS_Source

Connection Schema Parameters

Linked service *
LS_BlobStorage

Test connection Edit + New Learn more

File path *
raw-data

Compression type
No compression

Column delimiter
Comma (,)

Row delimiter

Properties

General Related

Name *
DS_Source

Description

Annotations
+ New

Cancel

Factory validation output

Your factory has been validated.
No errors were found.

Close

Microsoft Azure | Data Factory | adf-week4-arati

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Factory Resources

- Pipelines: 0
- Change Data Capture (preview): 0
- Datasets: 1
 - DS_Source
- Data flows: 0
- Power Query: 0

DS_Source

Column name	Type
Row ID	String
Order ID	String
Order Date	String
Ship Date	String
Ship Mode	String
Customer ID	String
Customer Name	String
Segment	String
Country	String
City	String
State	String
Postal Code	String

Properties

General

Name: DS_Source

Description:

Annotations: + New

Cancel Close

Publishing completed Successfully published

Your factory has been validated. No errors were found.

Microsoft Azure | Data Factory | adf-week4-arati

Storage Account

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Factory Resources

- Pipelines: 0
- Change Data Capture (preview): 0
- Datasets: 2
 - DS_Destination
 - DS_Source
- Data flows: 0
- Power Query: 0

DS_Destination

DelimitedText CSV

Properties

General

Name: DS_Destination

Description:

Annotations: + New

Cancel Close

Publishing completed Successfully published

Your factory has been validated. No errors were found.

Connection

Linked service: LS_BlobStorage

File path: processed-data

Compression type: No compression

Column delimiter: Comma (,)

Row delimiter:

Microsoft Azure | Data Factory | adf-week4-arati

Storage Account

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Factory Resources

- Pipelines: 1
 - PL_Copy_Superstore
- Change Data Capture (preview): 0
- Datasets: 2
 - DS_Destination
 - DS_Source
- Data flows: 0
- Power Query: 0

PL_Copy_Superstore

Get Metadata

Copy data

Copy data1

Properties

General

Name: PL_Copy_Superstore

Description:

Annotations: + New

Cancel Close

Factory validation output

Your factory has been validated. No errors were found.

General

Sink dataset: DS_Destination

Copy behavior: Select...

Max concurrent connections:

Task 4: Build the Pipeline with the Copy Data Activity

A new pipeline named **PL_Copy_Superstore** was created to automate the data transfer process.

The pipeline consisted of two activities:

- **Get Metadata** – Validates the source file.
- **Copy Data** – Copies the CSV file from the **raw-data** container to the **processed-data** container.

Both activities were connected so that the Copy Data activity started only after the Get Metadata activity completed successfully.

Since only one CSV file was used, a **ForEach** activity was not required.

This screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists 'Pipelines' (1) and 'Datasets' (2). The 'PL_Copy_Superstore' pipeline is selected. The main canvas displays a visual workflow with two activities: 'Get Metadata1' and 'Copy data1'. The 'Copy data1' activity is selected, and its 'Sink' tab is active, showing 'DS_Destination' as the sink dataset. The 'Properties' pane on the right shows the activity's name as 'PL_Copy_Superstore'. The 'Factory validation output' pane on the far right displays a success message: 'Your factory has been validated. No errors were found.'

This screenshot shows the Microsoft Azure Data Factory interface with the 'Get Metadata' activity selected. The 'Settings' tab is active, showing 'DS_Source' as the dataset. The 'Field list' section shows a list of fields: 'Argument', 'Exists', 'Size', and 'Last modified'. The 'Properties' pane on the right shows the activity's name as 'PL_Copy_Superstore'. The 'Factory validation output' pane on the far right displays a success message: 'Your factory has been validated. No errors were found.'

Microsoft Azure | Data Factory | adf-week4-arati

Storage Account

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»

Data Factory

Validate all

Publish all

Factory search

DS_Source

DS_Destination

PL_Copy_Superstore

Migrate to Fabric (Preview)

Factory Resources

Filter resources by name

Pipelines1

PL_Copy_Superstore

Change Data Capture (preview)0

Datasets2

DS_Destination

DS_Source

Data flows0

Power Query0

»

Validate

Debug

Add trigger

Get Metadata

Get Metadata1

Parameters

Variables

Settings

Output

Pipeline run ID

909c8d5c-0e06-468d-9d73-fc84659ae5d4

ⓘ

⌂

⌂ Pipeline status

✔ Succeeded

View debug run consumption

All status

Showing 1 - 2 of 2 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runti
Copy data1	✔ Succeeded	Copy data	7/12/2026, 1:59:04 AM	15s	AutoResolveInteg
Get Metadata1	✔ Succeeded	Get Metadata	7/12/2026, 1:58:49 AM	14s	AutoResolveInteg

Properties

General

Related

Name *

PL_Copy_Superstore

Description

Annotations

+ New

Cancel

Task 5: Execute and Monitor the Pipeline

After completing the pipeline configuration, I validated the pipeline and published all changes. I then executed the pipeline using the **Debug** option.

The pipeline ran successfully, and both the **Get Metadata** and **Copy Data** activities completed without any errors.

I verified the execution from the **Monitor** section and confirmed that the **Superstore.csv** file had been copied successfully to the **processed-data** container.

Microsoft Azure | Data Factory | adf-week4-arati

Pipeline runs

Triggered Debug Rerun Cancel options Refresh Edit columns List Gantt

Filter by run ID or name Chennai, Kolkata, Mu... : Last 24 hours Pipeline name: All Status: All Add filter Copy filters Export to CSV

Showing 1 - 1 items

Pipeline name	Run start	Run end	Duration	Status	Triggered by	Run ID	Parameters
PL_Copy_Superstore	7/12/2026, 1:58:42 A	7/12/2026, 1:59:20 A	38s	Succeeded	Manual trigger	909c8d5c-0e06-468d-...	

Last refreshed 0 minutes ago

Microsoft Azure

processed-data

Container

Search

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

processed-data

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

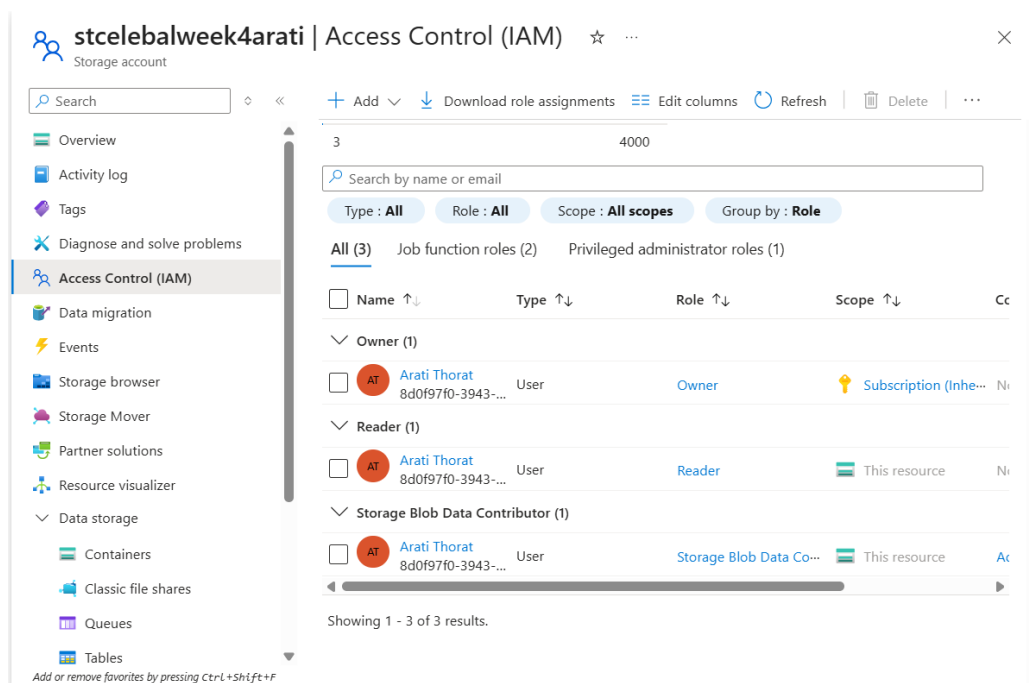
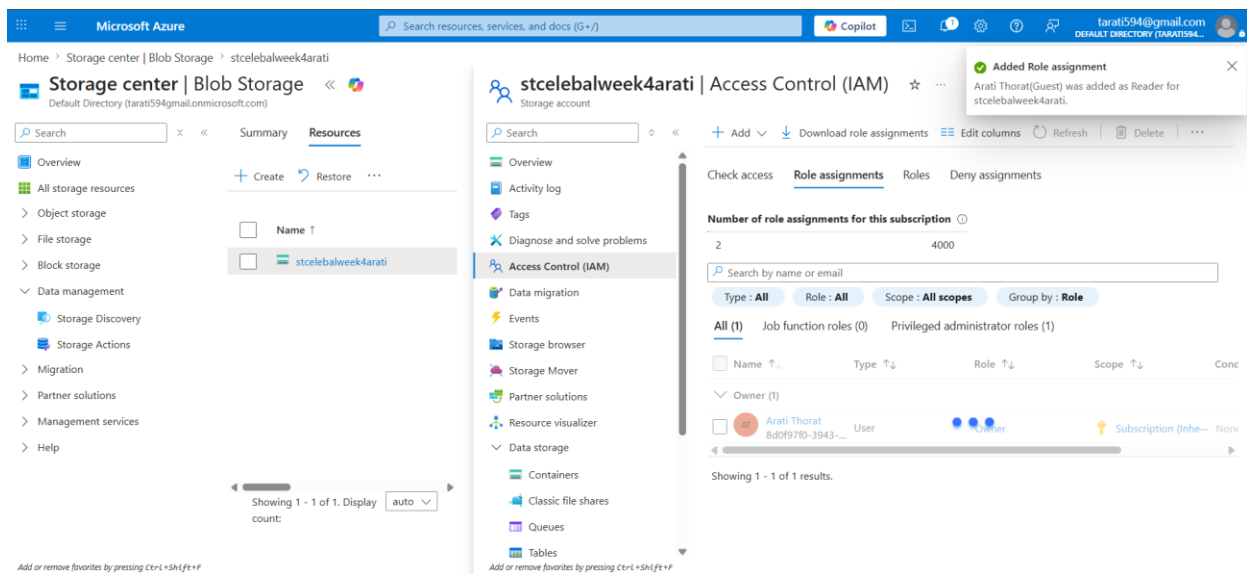
Only show active blobs

Showing all 1 items

Name	Last modified	Access tier	Blob type	Size	Lease state
Superstore.csv	7/12/2026, 1:59:16 AM	Hot (Inferred)	Block blob	2.18 MiB	Available

Task 6: Assign IAM Roles

- To understand Azure security, I explored **Identity and Access Management (IAM)**.
- For this assignment, I reviewed and configured the required roles to allow secure access between Azure Data Factory and the Storage Account.
- The following Azure roles were studied:
- **Reader** – Allows users to view Azure resources.
- **Contributor** – Allows users to manage Azure resources.
- **Storage Blob Data Contributor** – Allows Azure Data Factory to access Blob Storage securely.
- Understanding these roles helped me learn how Azure controls resource access using Role-Based Access Control (RBAC).



Mini Project: End-to-End Data Pipeline

The final mini project involved building a complete data pipeline using Azure Data Factory. The objective was to read a CSV file from Azure Blob Storage, validate it, and copy it to another location within the same storage account.

The pipeline followed these steps:

1. The **Superstore.csv** file was stored in the **raw-data** container.
2. The **Get Metadata** activity checked whether the file existed and retrieved basic information such as the file name, size, and last modified date.
3. After successful validation, the **Copy Data** activity copied the file from the **raw-data** container to the **processed-data** container.
4. The copied file was successfully created in the destination container, confirming that the pipeline executed correctly.

This mini project helped me understand how different Azure Data Factory components work together to build an automated data movement process. It also gave me practical experience in connecting Azure Storage with ADF, validating source data, executing pipelines, and verifying the final output.

Result: The pipeline executed successfully, the metadata was validated, and the CSV file was copied to the destination container without any errors.

The screenshot displays the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane shows a list of resources including Pipelines (1), Datasets (2), and Data flows (0). The 'PL_Copy_Superstore' pipeline is selected. The main workspace shows the pipeline design with two activities: 'Get Metadata' and 'Copy data'. The 'Get Metadata' activity is highlighted, and its 'Output' tab is active, displaying the following table:

Activity name	Activity status	Activity name	Run start	Duration	Integration runtime
Copy data1	✓ Succeeded	Copy data	7/12/2026, 1:59:04 AM	15s	AutoResolveInteg
Get Metadata1	✓ Succeeded	Get Metadata	7/12/2026, 1:58:49 AM	14s	AutoResolveInteg

The 'Pipeline status' is 'Succeeded'. The 'Properties' pane on the right shows the pipeline name 'PL_Copy_Superstore' and a description field.

Microsoft Azure | Data Factory | adf-week4-arati | Storage Account

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Factory validation output

Factory Resources

- Pipelines: 1
 - PL_Copy_Superstore
- Change Data Capture (preview): 0
- Datasets: 2
 - DS_Destination
 - DS_Source
- Data flows: 0
- Power Query: 0

Validate | Validate copy runtime | Debug | Add trigger

Get Metadata

Copy data

Copy data1

Properties

General

Name *

PL_Copy_Superstore

Description

Annotations

+ New

Cancel

Close

Microsoft Azure | Data Factory | adf-week4-arati | Search

Factory validation output

Factory Resources

- Pipelines: 1
 - PL_Copy_Superstore
- Change Data Capture (preview): 0
- Datasets: 2
 - DS_Destination
 - DS_Source
- Data flows: 0
- Power Query: 0

DS_Destination

DelimitedText

DS_Destination

Connection

Linked service *

LS_BlobStorage

File path *

processed-data / Directory / Superstore.csv

Compression type

No compression

Column delimiter

Comma (,)

Row delimiter

Default (\r\n, or \n)

Encoding

Default(UTF-8)

Quote character

Double quote (")

Test connection | Edit | + New | Learn more

Close

• Observations

- ADF During this assignment, I gained practical experience working with Azure Data Factory and Azure Blob Storage. Some of my observations are:
- At first, Azure Data Factory looked complex, but after understanding the relationship between **Linked Service, Dataset, Activity, and Pipeline**, it became much easier to work with.
- Creating a **Linked Service** only establishes the connection to the storage account, while **Datasets** define the source and destination files used in the pipeline.
- The **Get Metadata** activity proved useful because it checks whether the source file is available before the copy operation starts, reducing the chances of pipeline failure.
- Running the pipeline using **Debug** helped me test the workflow before publishing it.
- Using meaningful names such as **LS_BlobStorage, DS_Source, DS_Destination, and PL_Copy_Superstore** made the project easier to understand and manage.

Key Learnings

- This assignment helped me understand the basic concepts of Microsoft Azure and Azure Data Factory. The key learnings are:
- Learned how Azure resources are organized using **Subscriptions, Resource Groups, and Services**.
- Understood how **Azure Blob Storage** is used to store and manage files in the cloud.
- Learned to create and configure **Linked Services, Datasets, and Pipelines** in Azure Data Factory.
- Understood the purpose of **Get Metadata** for validating the source file before copying data.
- Gained hands-on experience with the **Copy Data** activity to move files between Blob Storage containers.
- Learned the difference between **Debug** and **Trigger** execution and how to monitor pipeline runs.
- Explored **Azure IAM** and understood how Role-Based Access Control (RBAC) is used to manage access to Azure resources.

Conclusion

This assignment provided me with practical experience in Microsoft Azure and Azure Data Factory. I successfully created a complete data pipeline that copied a CSV file from the **raw-data** container to the **processed-data** container after validating the source file using the **Get Metadata** activity.

Through this project, I gained a better understanding of Azure Storage, Azure Data Factory, datasets, linked services, pipelines, and cloud-based data movement. I also became familiar with Azure IAM and the basic concepts of access management. Overall, this assignment improved my confidence in working with Azure services and provided a strong foundation for learning more advanced data engineering concepts in future internship tasks.